

Reinforcement Learning

Reinforcement Learning is a Machine Learning approach where an agent learns by interacting with an environment. The agent performs actions and receives feedback.

The main concepts are:

- Agent The system making decisions.
- Environment The world or system the agent interacts with.
- Action A decision the agent can make.
- Reward Feedback indicating whether an action was beneficial.

The goal of the agent is to learn a strategy that maximizes its total reward over time.

For example, imagine an AI learning to play a simple game. The AI may receive:

- +10 points for winning
- +1 point for achieving an intermediate objective
- -10 points for losing

Initially, the agent may make poor decisions. After repeatedly interacting with the environment and receiving rewards or penalties, it can learn which actions are more successful.

Reinforcement Learning is different from supervised learning because the agent is not given the correct answer for every situation.